BELAYNEW ZEWDIE

Full-Stack | DevOps Engineer

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PROFESSIONAL SUMMARY

Dynamic and solution-oriented Electronics and Communication Engineering graduate transitioning into a high-performance Full-Stack and DevOps Engineer. Proven track record of architecting end-to-end applications using the MERN/PERN Stacks, Next.js, and WordPress, while mastering containerized workflows via Docker and Linux administration. Experienced in embedded IoT systems and applied computer vision, with a growing focus on AI-assisted product development. Certified in Artificial Intelligence Foundations, specializing in building resilient, production-ready systems tailored to local infrastructural challenges, such as network-resilient hybrid-cloud synchronization systems.

CORE COMPETENCIES

- Full-Stack Development: JavaScript (ES6+), React.js, Vite, Next.js, Node.js, Express.js, NestJS, WordPress (Astra/Elementor).
- Authentication & Security: JWT, OAuth/Google Sign-In, email OTP verification, role-based access control, protected routes and admin authorization.
- DevOps & Infrastructure: Docker (Containerization), Linux CLI, AWS Cloud Services, Server Environment Setup, event-driven/queue-based architectures.
- Databases & Tools: PostgreSQL (Prisma ORM), MongoDB, Supabase, Git/GitHub, Tailwind CSS v4, Socket.IO, REST APIs.
- Embedded Systems & IoT: Arduino, ESP8266, sensor integration, ThingSpeak, Embedded C/C++.
- Emerging Tech: Foundations of Artificial Intelligence, AI-assisted rapid prototyping, prompt engineering, vector/semantic search, computer vision (YOLO).

KEY PROJECTS

Hybrid-Cloud Edge Synchronization Engine
Role: Lead Architect & Developer

- Designed and engineered an enterprise-level resilience synchronization system tailored to maintain data integrity and application uptime during unexpected internet outages.
- Implemented localized edge-caching mechanisms that seamlessly synchronize local database states back to central cloud environments once connectivity is re-established.
- Built on NestJS and PostgreSQL with conflict detection, event replay, idempotency, and queue/outbox patterns for high-availability operation.

Merkato Store – Full-Stack E-Commerce Platform
Role: Full-Stack Engineer

- Architected a high-performance management and ordering ecosystem built entirely on the MERN Stack (MongoDB, Express, React, Node.js).
- Developed a robust e-commerce application designed to digitize local retail operations, featuring optimized performance and seamless user experiences.
- Engineered secure checkout logic, an interactive order tracking system (Placed → Processed → Shipped → Delivered) with email notifications, and reliable payment gateway integrations.
- Integrated an AI-powered chatbot and product search feature, using natural-language and vector-based search concepts to improve product discovery.
- Utilized containerized environments for reliable deployment across local and cloud servers.

Hotel Management, Food Ordering & Kitchen/Inventory System
Role: Full-Stack & Database Developer

- Architected a high-performance management and ordering ecosystem built entirely on the PERN Stack (PostgreSQL, Express, React, Node.js).

- Built a QR code-based smart ordering flow: customers scan a table QR code, browse the digital menu, place orders, and track order progress in real time.
- Designed a relational database schema ensuring strict data integrity for tracking real-time room bookings, food menu adjustments, and customer ordering lifecycles.
- Developed integrated Admin, Kitchen, and Store/Inventory modules covering recipes, ingredients, cooking status, automatic stock deduction, waste recording, and audit information.
- Built a highly responsive administrative dashboard allowing managers to track sales trends and live kitchen orders instantly.

IoT-Based Patient Monitoring System

Role: Embedded Systems Developer | Achievement: Best Project of the Year – 2025

- Developed an IoT-based patient monitoring system measuring blood pressure and body temperature using embedded sensors and microcontrollers.
- Transmitted patient health data to an online monitoring platform (ThingSpeak), allowing healthcare professionals to remotely monitor patients.
- Built with ESP8266, Arduino, and Embedded C++ for reliable sensor data acquisition and wireless transmission.

Video Watch & Earn Platform

Role: Full-Stack Developer (in development)

- Designed a platform where administrators publish YouTube videos and users earn coins by completing video-watching tasks.
- Built watch-hour verification, withdrawal requests, referrals, admin approval, payment tracking, and user/admin dashboards using the YouTube API/Player.

AI-Powered Ethiopian Job Marketplace

Role: Full-Stack Developer

Designed and developed an AI-powered recruitment platform connecting Ethiopian job seekers with companies and recruiters.

• Built job posting, job search, candidate profiles, CV management, job applications, application tracking, and user/admin dashboards.

• Implemented AI-powered job and candidate matching based on skills, experience, education, and job requirements.

Real-Time Expense Tracker Dashboard

Role: Frontend Developer

- Developed a modern, lightning-fast financial tracking application utilizing React.js, Vite, and the Tailwind CSS v4 framework.
- Optimized dashboard layouts for responsive visualization, ensuring data mutates accurately on the UI instantly.

Enterprise Real Estate Website

Role: CMS Developer

- Architected a complete property-listing web platform leveraging WordPress, customized via the Astra Theme and Elementor Pro.

EDUCATION

Bachelor of Science in Electronics and Communication Engineering

Adama Science and Technology University (ASTU) | 2021 – 2025, Ethiopia

CERTIFICATIONS

- Foundations of Artificial Intelligence – Youth Empowerment and Economic Accelerator Platform (YEEAP), in partnership with Florida University Southeast
- Certificate Course in Web Development – Udemy / Aakaro Consulting
- WordPress Web Development: Zero to Hero – Udemy
- Full Stack Web Development and Enterprise Application – Qiyas Web Development